

# Experiment - 02

Advanced Database Management Systems — Lab Record

|                      |               |                             |               |
|----------------------|---------------|-----------------------------|---------------|
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## (A) Average Marks by Department

A university administration requires an analytical performance report for each academic unit. Write a PostgreSQL solution to calculate the average marks scored by students grouped by their respective departments. The solution must join the student profile, course subjects, and grade marks tables, returning a breakdown sorted by the highest average marks down to the lowest.

### Output

| Department       | Average Marks |
|------------------|---------------|
| Mathematics      | 95.00         |
| Computer Science | 83.33         |

### Solution

```
SELECT D.DEPT_NAME AS DEPARTMENT,  
       ROUND(AVG(M.MARKS_SCORED), 2) AS AVERAGE_MARKS  
FROM DEPARTMENTS AS D  
JOIN STUDENTS AS S ON D.DEPT_ID = S.DEPT_ID  
JOIN MARKS AS M ON S.STUDENT_ID = M.STUDENT_ID  
GROUP BY D.DEPT_NAME  
ORDER BY AVERAGE_MARKS DESC;
```

## (B) Merging HR Salary Records

Two legacy HR systems (A and B) have separate records of employee salaries. These records may overlap. Management wants to merge these datasets and identify each unique employee (by EmpID) along with their lowest recorded salary across both systems.

### Objective

- Combine two tables A and B.
- Return each EmpID with their lowest salary, and the corresponding Ename.

Table A

| EmpID | Ename | Salary |
|-------|-------|--------|
| 1     | AA    | 1000   |
| 2     | BB    | 300    |

Table B

| EmpID | Ename | Salary |
|-------|-------|--------|
| 2     | BB    | 400    |
| 3     | CC    | 100    |

### Output

| EmpID | Ename | Salary |
|-------|-------|--------|
| 1     | AA    | 1000   |
| 2     | BB    | 300    |
| 3     | CC    | 100    |

### Solution

```
SELECT EMPID, MIN(ENAME) AS ENAME, MIN(SALARY) AS SALARY
FROM (
    SELECT * FROM A
    UNION ALL
    SELECT * FROM B
) AS INTERMEDIATE_RES
GROUP BY EMPID;
```

## (C) Highest-Paid Employee per Department

A multinational organization maintains separate records for employee information and department details. The management wants to identify the employees who receive the highest salary within their respective departments for an annual performance report.

If more than one employee earns the same highest salary in a department, all such employees should be included in the report. The final output should display the Department Name, Employee Name, and Salary, arranged in ascending order of department name.

**Note:** The employee and department information are maintained in separate normalized tables.

Input Table: Employee

| ID | Name  | Salary | Dept_ID |
|----|-------|--------|---------|
| 1  | JOE   | 70000  | 1       |
| 2  | JIM   | 90000  | 1       |
| 3  | HENRY | 80000  | 2       |
| 4  | SAM   | 60000  | 2       |
| 4  | MAX   | 90000  | 1       |

Department

| ID | Dept_Name |
|----|-----------|
| 1  | IT        |
| 2  | SALES     |

### Output

| Dept_Name | Name  | Salary |
|-----------|-------|--------|
| IT        | MAX   | 90000  |
| IT        | JIM   | 90000  |
| SALES     | HENRY | 80000  |

**Part B: Find the 2<sup>nd</sup> highest earning employee from the same schema.**

### Solution

(a)

```
SELECT D.DEPT_NAME, E.NAME, E.SALARY
FROM EMPLOYEE AS E
JOIN DEPARTMENT AS D ON E.DEPARTMENT_ID = D.ID
WHERE E.SALARY IN (
    SELECT MAX(SALARY)
    FROM EMPLOYEE AS E1
    WHERE E1.DEPARTMENT_ID = E.DEPARTMENT_ID
)
ORDER BY DEPT_NAME;
```

(b)

```
SELECT D.DEPT_NAME, E.NAME, E.SALARY
FROM EMPLOYEE AS E
JOIN DEPARTMENT AS D ON E.DEPARTMENT_ID = D.ID
WHERE E.SALARY IN (
    SELECT MAX(SALARY)
    FROM EMPLOYEE AS E1
    WHERE E1.DEPARTMENT_ID = E.DEPARTMENT_ID
    AND E1.SALARY NOT IN (
        SELECT MAX(SALARY)
        FROM EMPLOYEE AS E2
        WHERE E2.DEPARTMENT_ID = E1.DEPARTMENT_ID
    )
)
ORDER BY DEPT_NAME;
```